

Practical - 5

Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem definition: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Theory:

1. Are correct input types and attributes used?

Answer: Yes. HTML5 input types such as text, email, tel, and password are used with suitable attributes like placeholder, maxlength, and required.

2. Are validation errors displayed near the relevant fields?

Answers: yes. JavaScript displays specific error messages directly below the corresponding input field.

3. Is password strength checked?

Answers: Yes. JavaScript and Regular Expression are used to check password requirements such as minimum length, uppercase, lowercase, and numbers.

4. Is the form accessible using keyboard and screen-reader-friendly labels?

Answers: Yes. Proper <label> elements are used for form fields, and standard HTML form controls can be accessed using the keyboard.

A. Implementation/ Output/Result (Keep only one option which is suitable to practical)

- Javascript methods

Method / Feature javaPurpose

getElementById()	Accesses form elements
querySelector()	Checks selected radio buttons
test()	Tests input using Regular Expressions
onsubmit	Calls validation when form is submitted
checked	Checks whether checkbox/radio is selected
innerHTML	Displays error and success message

- **HTML tags**

Tag	Purpose
<html>	Defines the HTML document
<head>	Contains page information and title
<title>	Sets the webpage title
<style>	Adds CSS styling to the form
<body>	Contains the visible webpage content
<form>	Creates the student registration form
<h1>	Displays the main heading
<h2>	Displays the form heading
<h3>	Displays the success message
<label>	Provides labels for form fields
<input>	Creates text, email, mobile, password, radio and checkbox fields
<select>	Creates the course dropdown
<option>	Provides course options
<p>	Displays validation error messages
<button>	Creates form buttons
<script>	Contains JavaScript validation code
 	Moves content to the next line

A. Evidence of Work (Screenshots)

Students cannot register if it doesn't matches the given conditions ...it will show error

StudentHub
Student Registration

Name:

Email:

Mobile:

Enter a valid mobile number.

Password:

Password needs 6 characters, uppercase, lowercase and number.

Confirm Password:

Passwords do not match.

Course:

Year: ☐ 1st ☒ 2nd ☐ 3rd ☐ 4th

Gender: ☐ Male ☒ Female ☐ Other

☒ I accept Terms and Conditions

StudentHub

Student Registration

Name:

Email:

Mobile:

Password:

Confirm Password:

Course:

Year:
☐ 1st ☒ 2nd ☐ 3rd ☐ 4th

Gender:
☐ Male ☒ Female ☐ Other

☒ I accept Terms and Conditions

Registration Successful!

StudentHub

Student Registration

Name:

Email:

Mobile:

Password:

Confirm Password:

Course:

Year:
☐ 1st ☐ 2nd ☐ 3rd ☐ 4th

Gender:
☐ Male ☐ Female ☐ Other

☐ I accept Terms and Conditions

Registration Successful!

Clicking on reset , gives these.

A. GitHub Repository Link:

<https://github.com/diyatolia1212/StudentHub/tree/afcda0cfaf4fce41e85a1cc51e30f0b14dbccf21/PRACTICAL%205>

B. Challenges Faced:

- Implementing JavaScript validation for all form inputs.
- Checking password strength and matching passwords.
- Displaying error messages near the correct fields.
- Handling radio buttons and checkbox validation.

5Learning Outcomes:

- Learned to create student registration forms using HTML5.
- Learned to validate user input using JavaScript and Regular Expressions (Regex).
- Learned to implement password strength and password matching checks.
- Learned to display user-friendly validation error messages.

Faculty Remarks: (Keep this blank for faculty)